



Yakeen Plus (2025)

Practice Test - 16

DURATION : 200 Minutes

DATE : 19/12/2024

M. MARKS : 720

Topics Covered

Physics :	Electrostatics, Current Electricity, Magnetic Effects of Current and Magnetism, Electromagnetic Induction and Alternating Currents, Electromagnetic Waves
Chemistry :	Thermodynamics & Thermochemistry, Chemical Kinetics, Alcohols, Ethers and Phenols, Aldehydes, Ketones and Carboxylic Acids, The p-Block Elements (Group 13,14,15, 16, 17, 18), The d and f-Block Elements
Biology :	(Botany): Sexual Reproduction in Flowering Plant, Principle of Inheritance and Variation, Microbes in Human Welfare (Zoology): Human Reproduction, Reproductive Health, Evolution

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are **four Section** in the Question Paper, **Section I, II, III & IV** consisting of Section-I (Physics from question number **1 to 50**), Section-II (Chemistry: from question number **51 to 100**), Section-III (Botany: from question number **101 to 150**) & Section IV (Zoology: from question number **151 to 200**), having 50 Questions in each Subject and each subject is divided in two Section, Section A consisting 35 questions (all questions are compulsory) and Section B consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

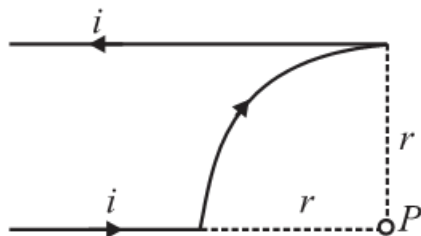
OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

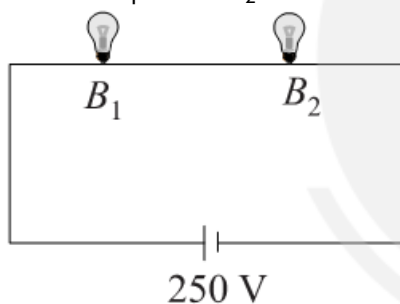
Practice Test-16

Yakeen Plus (2025)

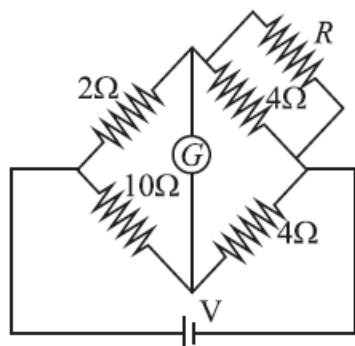
- Q1** A current carrying conductor is placed as shown in figure. The magnitude of magnetic field at point P will be:



- (1) $\frac{\mu_0}{4\pi} \frac{2i}{r} \left(\frac{\pi}{2} + 1 \right)$
 (2) $\frac{\mu_0}{4\pi} \frac{2i}{r} \left(\frac{\pi}{2} - 1 \right)$
 (3) $\frac{\mu_0}{4\pi} \frac{i}{r} \left(\frac{\pi}{2} + 1 \right)$
 (4) $\frac{\mu_0}{4\pi} \frac{i}{r} \left(\frac{\pi}{2} - 1 \right)$
- Q2** Bulb B_1 (100 W – 250 V) and bulb B_2 (100 W – 200 V) are connected across 250 V. What is potential drop across B_2 ?



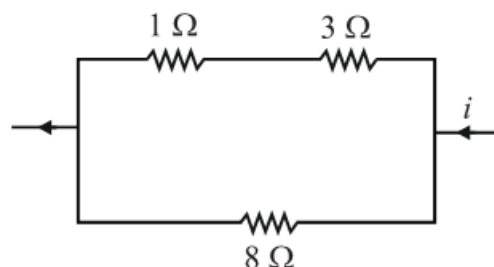
- (1) 200 V
 (2) 98 V
 (3) 250 V
 (4) 48 V
- Q3** The value of R for which deflection in galvanometer G is zero.



- (1) 2Ω
 (2) 1Ω
 (3) 4Ω
 (4) 8Ω

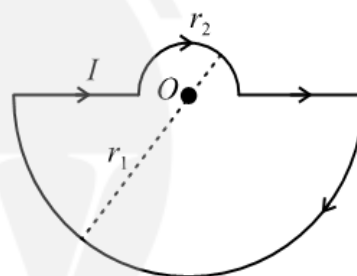
Q4

- Power dissipated across the 8Ω resistor in the circuit shown here is 2 W. The power dissipated in watt units across the 3Ω resistor is:

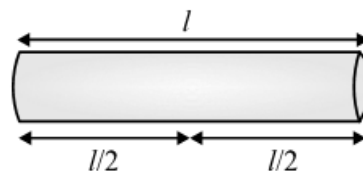


- (1) 2
 (2) 1
 (3) 0.5
 (4) 3

- Q5** In the given loop the magnetic field at the centre O is:



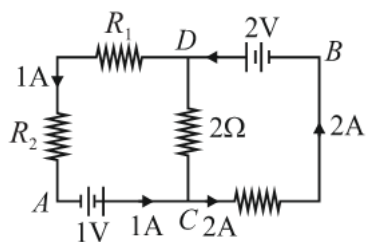
- (1) $\frac{\mu_0 I}{4} \left(\frac{r_1 + r_2}{r_1 r_2} \right)$ out of the page.
 (2) $\frac{\mu_0 I}{4} \left(\frac{r_1 + r_2}{r_1 r_2} \right)$ into the page.
 (3) $\frac{\mu_0 I}{4} \left(\frac{r_1 - r_2}{r_1 r_2} \right)$ out of the page.
 (4) $\frac{\mu_0 I}{4} \left(\frac{r_1 - r_2}{r_1 r_2} \right)$ into the page.
- Q6** A metal wire of length l has resistance R . Half of its length is stretched uniformly such that final length of wire becomes $2l$. The new resistance of wire will be?



- (1) $2R$
 (2) $3R$
 (3) $4R$
 (4) $5R$
- Q7** In the circuit shown in the figure, if the potential at point A is taken to be zero, the potential at

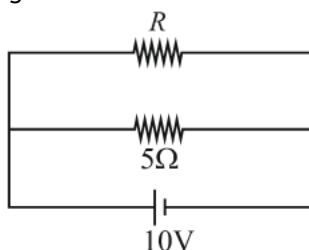


point B is:



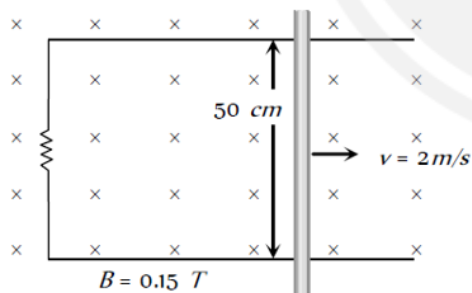
- (1) +1V (2) -1V
(3) +2V (4) -2V

Q8 The power dissipated in the circuit shown in the figure is 30 watts. The value of R is:



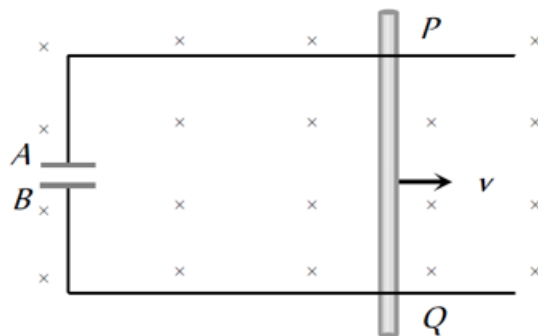
- (1) 10 Ω (2) 30 Ω
(3) 20 Ω (4) 15 Ω

Q9 As shown in the figure a metal rod makes contact and complete the circuit. The circuit is perpendicular to the uniform magnetic field with $B = 0.15$ tesla. If the resistance is 3 Ω , force needed to move the rod as indicated with a constant speed of 2 m/s is:



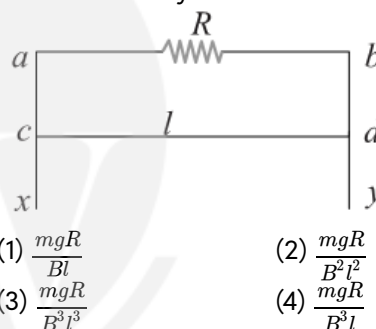
- (1) 3.75×10^{-3} N
(2) 3.75×10^{-2} N
(3) 3.75×10^2 N
(4) 3.75×10^{-4} N

Q10 A conducting rod PQ of length $L = 1.0$ m is moving with a uniform speed $v = 2$ m/s in a uniform magnetic field $B = 4.0$ T directed into the paper. A capacitor of capacity $C = 10$ μ F is connected as shown in figure. Then



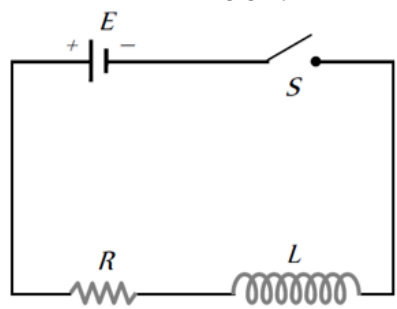
- (1) $q_A = +80$ μ C and $q_B = -80$ μ C
(2) $q_A = -80$ μ C and $q_B = +80$ μ C
(3) $q_A = 0 = q_B$
(4) $q_A = -40$ μ C and $q_B = +40$ μ C

Q11 A wire cd of length l and mass m is sliding without friction on conducting rails ax and by as shown. The vertical rails are connected to each other with a resistance R between a and b . A uniform magnetic field B is applied perpendicular to the plane $abcd$ such that cd moves with a constant velocity of:



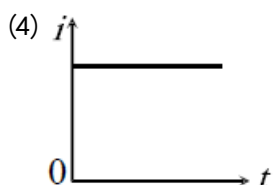
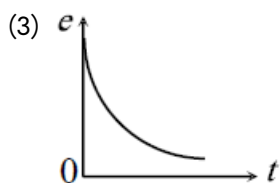
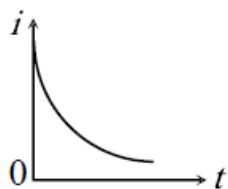
- (1) $\frac{mgR}{Bl}$ (2) $\frac{mgR}{B^2 l^2}$
(3) $\frac{mgR}{B^3 l^3}$ (4) $\frac{mgR}{B^3 l}$

Q12 Switch S of the circuit shown in figure, is closed at $t = 0$. If e denotes the induced emf in L and i , the current flowing through the circuit at time t , which of the following graphs is correct

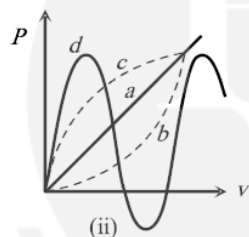
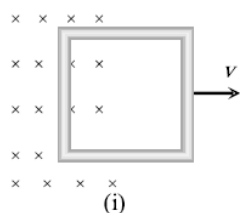


- (1)
(2)



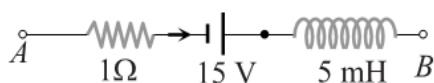


- Q13** Figure (i) shows a conducting loop being pulled out of a magnetic field with a speed v . Which of the four plots shown in figure (ii) may represent the power delivered by the pulling agent as a function of the speed v



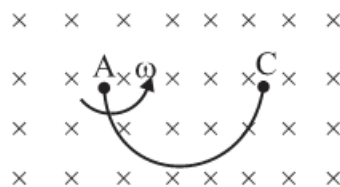
- (1) a (2) b
(3) c (4) d

- Q14** The network shown in the figure is a part of a complete circuit. If at a certain instant the current i is 5 A and is decreasing at the rate of 10^3 A/s then $V_A - V_B$ is



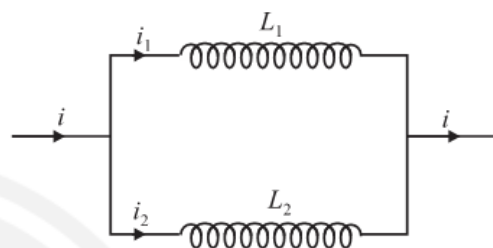
- (1) 5 V (2) 10 V
(3) 15 V (4) 20 V

- Q15** As shown in the figure, a semi-circular wire of radius R is rotated with angular speed ω about an axis perpendicular to the plane of wire and passing through A in the uniform magnetic field B_0 . Find the emf induced across wire



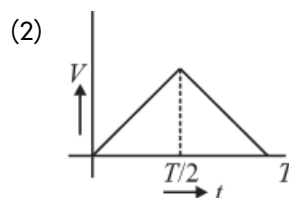
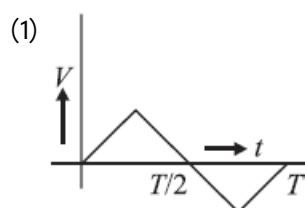
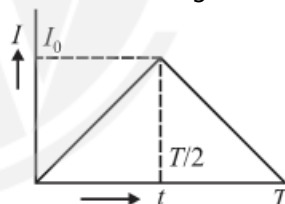
- (1) $\frac{1}{2} B_0 \omega R^2$ (2) $2 B_0 \omega R^2$
(3) $\frac{1}{2} B_0 R \omega^2$ (4) $B_0 R \omega^2$

- Q16** Two inductors L_1 and L_2 are connected in parallel and a time varying current flows as shown in figure. Then, the ratio of currents i_1/i_2 at any time t is:



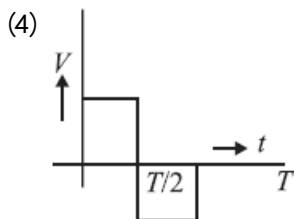
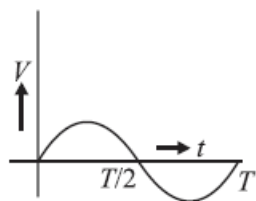
- (1) $\frac{L_1}{L_2}$ (2) $\frac{L_2}{L_1}$
(3) $\frac{L_1^2}{(L_1 + L_2)^2}$ (4) $\frac{L_2^2}{(L_1 + L_2)^2}$

- Q17** An alternating current i in an inductance coil varies with time according to the graph given below. Which one of the following graphs the variation of voltage with time?

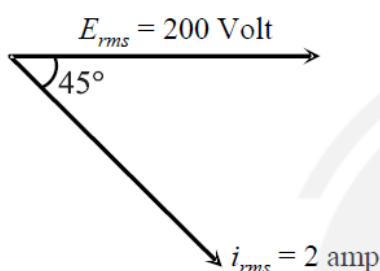


- (3)

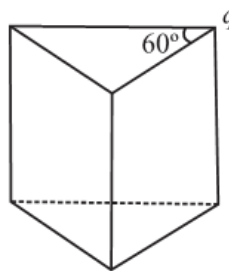




- Q18** The vector diagram of current and voltage for a circuit is as shown. The components of the circuit will be



- (1) LCR only
(2) LR only
(3) Either LCR or LR
(4) RC only
- Q19** Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $A = 10 \text{ cm}^2$ and length = 20 cm. If one of the solenoids has 300 turns and the other 400 turns, their mutual inductance is ($\mu_0 = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$)
- (1) $2.4\pi \times 10^{-5} \text{ H}$
(2) $4.8\pi \times 10^{-4} \text{ H}$
(3) $4.8\pi \times 10^{-5} \text{ H}$
(4) $2.4\pi \times 10^{-4} \text{ H}$
- Q20** A charge q is placed at a vertex of the given geometry of angle 60° . Then, the flux coming out from this geometry will be:

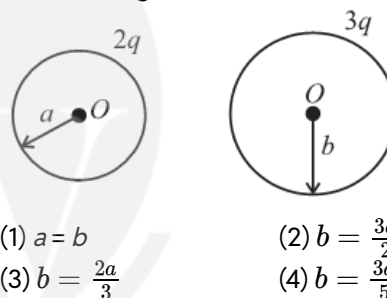


- (1) $\frac{q}{6\epsilon_0}$
(2) $\frac{q}{16\epsilon_0}$
(3) $\frac{q}{8\epsilon_0}$
(4) $\frac{q}{12\epsilon_0}$

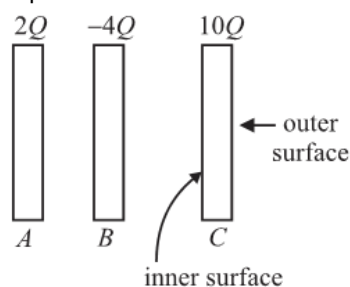
- Q21** The maximum electric field intensity on the axis of a uniformly charge ring of charge q and radius R will be:

- (1) $\frac{1}{4\pi\epsilon_0} \frac{q}{3\sqrt{3}R^2}$
(2) $\frac{1}{4\pi\epsilon_0} \frac{2q}{3R^2}$
(3) $\frac{1}{4\pi\epsilon_0} \frac{2q}{3\sqrt{3}R^2}$
(4) $\frac{1}{4\pi\epsilon_0} \frac{3q}{3\sqrt{3}R^2}$

- Q22** Two spheres are connected by a wire, there is no flow of charge between them if:



- Q23** Three very large metal plates are given charges as shown in the figure. If cross sectional area of each plate is same then final charge distribution on plate C is:

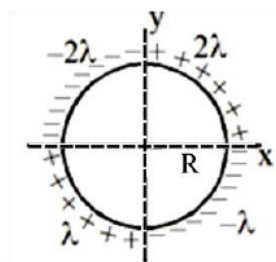


- (1) $+5Q$ on inner surface and $+5Q$ on outer surface
(2) $+6Q$ on inner surface and $+4Q$ on outer surface
(3) $+7Q$ on inner surface and $+9Q$ on outer surface
(4)

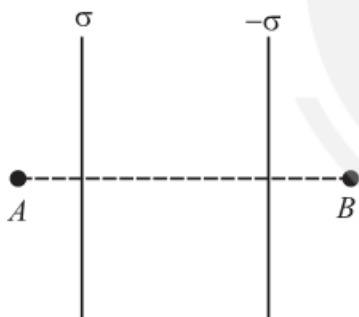


$+8Q$ on inner surface and $+2Q$ on outer surface

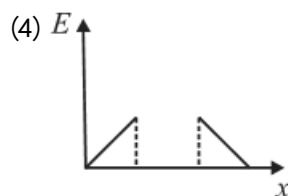
- Q24** The charge per unit length of the four quadrant of the ring is 2λ , -2λ , λ and $-\lambda$ respectively. The electric field at the centre is:



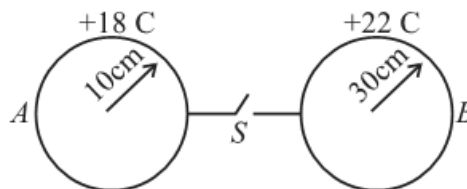
- (1) $-\frac{\lambda}{2\pi\epsilon_0 R} \hat{i}$
 (2) $-\frac{\lambda}{2\pi\epsilon_0 R} \hat{j}$
 (3) $\frac{\sqrt{2}\lambda}{2\pi\epsilon_0 R} \hat{i}$
 (4) $\frac{\sqrt{2}\lambda}{2\pi\epsilon_0 R} \hat{j}$
- Q25** Two large parallel planes charged uniformly with surface charge density σ and $-\sigma$ are located as shown in the figure. Which one of the following graphs shows the variation of electric field along a line perpendicular to the planes as one moves from A to B?



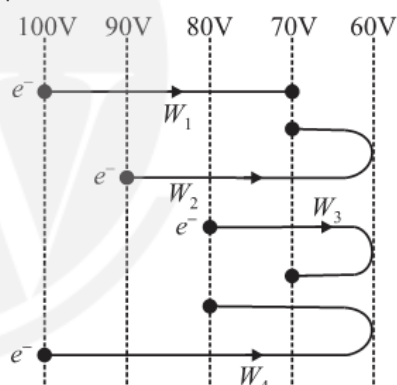
- (1)
- (2)
- (3)



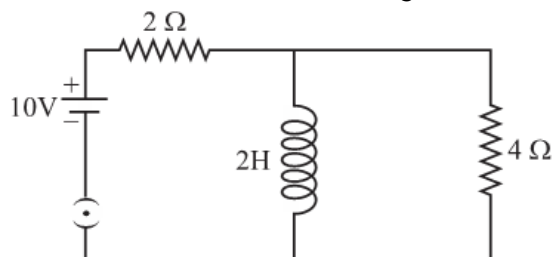
- Q26** On closing the switch, charge flowing through the wire will be:



- (1) $+8\text{C}$ from A to B
 (2) $+10\text{C}$ from B to A
 (3) -7C from B to A
 (4) -15C from B to A
- Q27** Figure shows a family of equipotential lines between which an electron is moved as shown in figure. Then **correct** order of work done in the process is:



- (1) $W_1 > W_2 > W_3 > W_4$
 (2) $W_1 < W_2 < W_3 < W_4$
 (3) $W_1 > W_2 = W_3 > W_4$
 (4) $W_1 > W_2 = W_4 > W_3$
- Q28** Current in 4Ω resistance shown in figure is:



- (a) Just after the closing of the key
 (b) Sometime after the closing of the key
 (1)



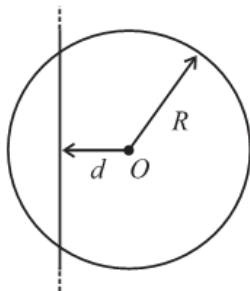
$$5A, \frac{5}{3}A$$

$$(2) \frac{5}{3}A, \text{ Zero}$$

$$(3) \frac{5}{3}A, 5A$$

$$(4) \text{ Zero}, 5A$$

- Q29** Calculate the electric flux through the Gaussian spherical surface of radius R , when a infinitely long line charge of linear charges density (λ), lies on it, at a distance d from its centre:



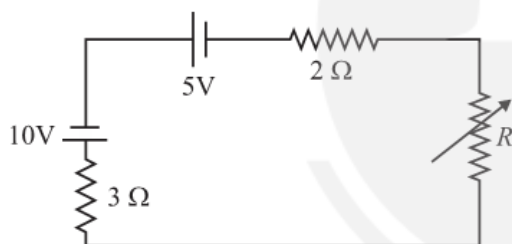
$$(1) \frac{\lambda \sqrt{R^2 - 2d^2}}{\epsilon_0}$$

$$(2) \frac{2\lambda \sqrt{R^2 - d^2}}{\epsilon_0}$$

$$(3) \frac{\lambda \sqrt{R^2 - 4d^2}}{\epsilon_0}$$

$$(4) \frac{2\lambda \sqrt{R^2 + d^2}}{\epsilon_0}$$

Q30



Find maximum power in resistance R :

$$(1) 11.25 \text{ W}$$

$$(2) 20 \text{ W}$$

$$(3) 15.5 \text{ W}$$

$$(4) 25 \text{ W}$$

- Q31** In the given diagrams, some arrangements of charges are formed by placing them on the vertices of some regular polygons in list-I (point O is the circumcenter of the given regular polygons in each diagram). The direction of net electric field at point O is shown in list-II. If nature of " Q " is positive, then choose the **correct** option by matching the **list-I** with the **list-II**

List-I		List-II	
(A)		(I)	
(B)		(II)	
(C)		(III)	
(D)		(IV)	

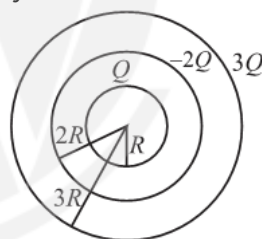
$$(1) \text{ A-III, B-I, C-II, D-IV}$$

$$(2) \text{ A-IV, B-III, C-II, D-I}$$

$$(3) \text{ A-II, B-III, C-IV, D-I}$$

$$(4) \text{ A-II, B-I, C-III, D-IV}$$

- Q32** Three metallic spherical shells of radius R , $2R$ and $3R$ has charges Q , $-2Q$ and $3Q$ respectively. If middle shell and outer most shell are connected by wire then what will be charge in middle shell.



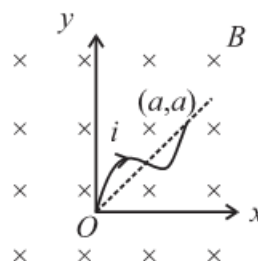
$$(1) \text{ Zero}$$

$$(2) Q$$

$$(3) -Q$$

$$(4) 3Q$$

- Q33** Find magnitude of magnetic force on given i current carrying wire placed in uniform magnetic field B .



$$(1) i a B$$

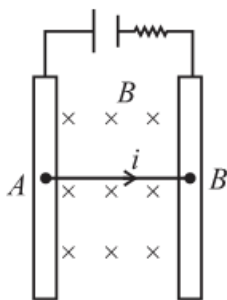
$$(2) 2 i a B$$

$$(3) \text{ zero}$$

$$(4) \sqrt{2} i a B$$

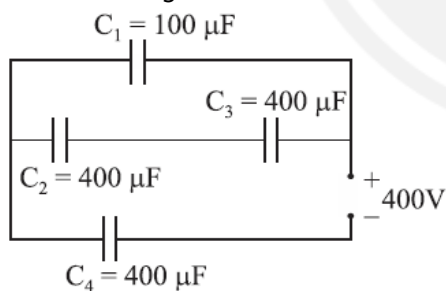


- Q34** A rod AB of mass m is hanged freely on two fixed conduction rail having distance l between them in a uniform magnetic field B as shown in figure. Find the current i so that rod AB is at rest.



- (1) $\frac{2mg}{lB}$ (2) $\frac{2mgl}{B}$
 (3) $\frac{2mgB}{l}$ (4) $\frac{mg}{lB}$
- Q35** Consider below statements:
 (i) In EMW both electric and magnetic fields are present.
 (II) Energy is higher in electric field than magnetic field.
 (III) Both electric and magnetic fields have same energy.
 False statement(s) is/are
 (1) (I), (II) (2) Only (II)
 (3) (I), (III) (4) Only (III)

- Q36** The equivalent capacitance for the network shown in the figure:



- (1) $\frac{1200}{7} \mu F$ (2) $\frac{1000}{4} \mu F$
 (3) $\frac{1800}{7} \mu F$ (4) $\frac{1300}{3} \mu F$
- Q37** A toroid has 6000 turns and inner and outer radius of 5 cm and 15 cm. It has soft iron core of relative permeability 1000. Find magnetic induction in the core when a current of 1A is passed through the toroid.
 (1) 9 T (2) 16 T
 (3) 12 T (4) 24 T

Q38

Magnetic field at the center of a circular coil having N turns carrying current is B . If A is the area of coil, the magnetic dipole moment of the coil is:

- (1) $\frac{NBA\sqrt{A}}{\mu_0\pi}$ (2) $\frac{BA\sqrt{A}}{\mu_0\pi}$
 (3) $\frac{2NBA}{\mu_0} \sqrt{\frac{A}{\pi}}$ (4) $\frac{2BA}{\mu_0} \sqrt{\frac{A}{\pi}}$

- Q39** A wire of given length is bent in the form of a ring of diameter $2a$ having self inductance (L), then L will depend upon following

- (1) $L \propto a^{-1}$
 (2) $L \propto a$
 (3) $L \propto a^{-2}$
 (4) $L \propto a^2$

- Q40** The quality factor of series L - C - R circuit is Q , then the value of capacitance in terms of resistance R and inductance (L) is:

- (1) $\frac{L^2}{R^2 Q^2}$ (2) $\frac{L}{(RQ)^2}$
 (3) $\frac{R^2 Q^2}{L^2}$ (4) $\frac{(RQ)^2}{L}$

- Q41** While keeping area of cross-section of a solenoid same, the number of turns and length of solenoid are both doubled. The self-inductance of the coil will be:

- (1) halved
 (2) doubled
 (3) $1/4$ times the original value
 (4) unaffected

- Q42** A wire having mass m and length l is at rest initially and can freely slide on a pair of parallel smooth horizontal rails placed in vertical magnetic field B . The rails are connected by a capacitor of capacitance C . The electric resistance of the rails and the wire is zero. If a constant force F acts on the wire as shown in the figure. Then, the acceleration of the wire can be given as

- (1) $\frac{C^2 B^2 l - F}{m + CBl}$ (2) $\frac{F}{m + CBl}$
 (3) $\frac{FC^2 B^2 l}{m}$ (4) $\frac{F}{m + CB^2 l^2}$

- Q43** A closely wound solenoid of 2000 turns and area of cross-section $1.5 \times 10^{-4} \text{ m}^2$ carries a current of 2.0 A. It is suspended through its centre and perpendicular to its length, allowing it to turn in a horizontal plane in a uniform



magnetic field 5×10^{-2} T, making an angle of 30° with the axis of the solenoid. The torque on the solenoid will be

- (1) 3×10^{-3} N-m
- (2) 1.5×10^{-3} N-m
- (3) 1.5×10^{-2} N-m
- (4) 3×10^{-2} N-m

Q44 A plane electromagnetic wave $E_z = 100 \cos(6 \times 10^8 t + 4x)$ V/m propagates in a medium of refractive index:

- (1) 1.5
- (2) 2.0
- (3) 2.4
- (4) 4.0

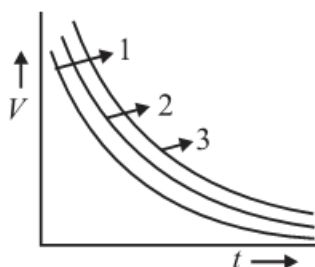
Q45 In a series R - L circuit, 1 V DC supply is given, if $R = 2\Omega$ and $L = 4$ H, then the value of current in 2 s is given by:

- (1) $\frac{3}{2}(1 - e^{-1})$
- (2) $\frac{1}{2}(1 - e)$
- (3) $\frac{3}{2}(1 + e)$
- (4) $\frac{1}{2}(1 - e^{-1})$

Q46 A series R - C circuit is connected to AC voltage source. Consider two cases; (A) When C is without a dielectric medium and (B) When C is filled with dielectric of dielectric constant 4. The current I_R through the resistor and voltage V_C across the capacitor are compared in two cases. Which of the following is **true**?

- (1) $I_R^A > I_R^B$
- (2) $I_R^A < I_R^B$
- (3) $V_C^A < V_C^B$
- (4) None of these

Q47 Three identical capacitors A , B and C are charged to the same potential and then made to discharge through three resistance $R_A > R_B > R_C$. Their potential differences (V) are plotted against time t , giving the curves 1, 2 and 3. Choose the **incorrect** option.

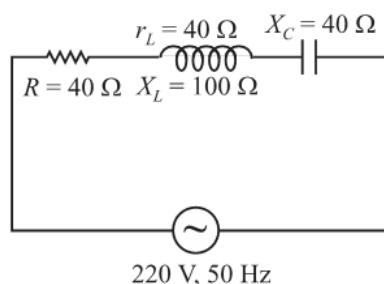


- (1) $1 \rightarrow A$
- (2) $2 \rightarrow B$
- (3) $1 \rightarrow C$
- (4) $3 \rightarrow A$

Q48 An alternating voltage $V = V_0 \sin \omega t$ is applied across a circuit. As result, $I = I_0 \sin(\omega t - \pi/2)$ flows in it. The power consumed per cycle is:

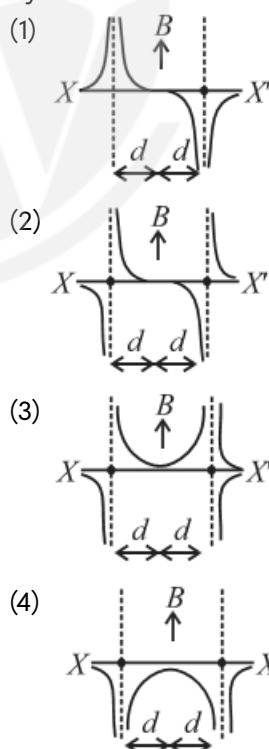
- (1) zero
- (2) $0.5 V_0 I_0$
- (3) $0.707 V_0 I_0$
- (4) $1.414 V_0 I_0$

Q49 The power factor of the circuit shown in figure is:



- (1) 0.2
- (2) 0.4
- (3) 0.8
- (4) 0.6

Q50 Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper, as shown. The variation of the magnetic field B along the line XX' is given by:



Q51 In the reversible adiabatic expansion of an ideal monoatomic gas, the final volume is 8 times the



initial volume. The ratio of final temperature to initial temperature is:

- (1) 8 : 1 (2) 1 : 4
(3) 1 : 2 (4) 4 : 1

Q52 Mechanism of a hypothetical reaction is given below;

$X_2 + Y_2 \rightarrow 2XY$ is given below;

- I. $X_2 \rightleftharpoons X + X$ (fast)
II. $X + Y_2 \rightarrow XY + Y$ (slow)
III. $X + Y \rightarrow XY$ (fast)

The overall order of reaction will be:

- (1) 1 (2) 2
(3) 0 (4) 1.5

Q53 Correct structure of IUPAC name of 2-Methoxy-2-methylpropane is:

- (1) $CH_3-CH_2-CH_2-CH_2-O-CH_3$
(2) $CH_3-CH_2-\underset{\substack{| \\ CH_3}}{CH}-O-CH_3$
(3) $\begin{array}{c} CH_3 \\ | \\ CH_3-C-O-CH_3 \\ | \\ CH_3 \end{array}$
(4) $CH_3-\underset{\substack{| \\ CH_3}}{CH}-CH_2-O-CH_3$

Q54 $KMnO_4$ on reaction with KOH does **not** give:

- (1) MnO_2
(2) K_2MnO_4
(3) H_2O
(4) Both (1) and (2)

Q55 Match the **List-I** with **List-II** and mark the appropriate choice.

List-I		List-II	
(A)	An element which can show +8 oxidation state	(I)	Ce
(B)	Most stable oxidation state of Mn in oxides	(II)	Pm
(C)	Radioactive lanthanoid	(III)	Os
(D)	Lanthanoid which shows +4 oxidation state	(IV)	+7

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-II, D-I

Q56 Number of carbon atoms present in malonic acid is:

- (1) 2 (2) 3
(3) 4 (4) 5

Q57 The ionization enthalpy of Pb is slightly greater than that of Sn due to:

- (1) increase in nuclear charge
(2) inert-pair effect
(3) shielding effect
(4) All of these.

Q58 Given below are two statements:

Statement I: large difference in boiling points of alcohols and ethers is due to the presence of hydrogen bonding in alcohols.

Statement II: Oxygen of ether can also form hydrogen bonding with water molecules.

In the light of above statements, Choose the **correct** option given below:

- (1) Both statement I and Statement II are correct.
(2) Both statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

Q59 Lanthanides and actinides resemble in:

- (1) electronic configuration
(2) oxidation state
(3) filling of $(n-2)f$ orbitals.
(4) both (2) and (3).

Q60 In the Arrhenius equation, the factor $e^{-E_a/RT}$ corresponds to:

- (1) the fraction of molecules that have kinetic energy less than E_a .
(2) the fraction of molecules that have kinetic energy greater than E_a .
(3) the fraction of molecules that have kinetic energy equal to zero.
(4) None of these

Q61 The extent of actinoid contraction is greater than lanthanoid contraction due to:

- (1) poor screening of $5f$ -electrons than $4f$ -electrons
(2)



poor screening of 4*f*-electrons than 5*f*-electrons

(3) poor screening of 3*d*-electrons than 5*f*-electrons

(4) None of these

Q62 Ozonolysis of C_7H_{14} gives 2-methylpentan-3-one Compound C_7H_{14} is:

(1) 2-Ethyl-3-methylbut-1-ene

(2) 3-Ethyl-2-methylbut-3-ene

(3) Hept-1-ene.

(4) Hept-2-ene.

Q63 Among the following, molecule has linear shape is:

(1) $XeOF_4$

(2) $XeOF_2$

(3) XeO_3

(4) XeF_2

Q64 At 298 K, the standard enthalpy change (ΔH°) for the neutralisation of HCN and NaOH is -2460 cal. Given that the standard enthalpy change for the neutralization of HCl and NaOH is -13.71 kcal. heat of ionization of HCN is:

(1) $+15820$ cal

(2) -11250 cal

(3) $+11250$ cal

(4) -15820 cal

Q65 Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Hydrolysis of cane sugar is a pseudo first order reaction.

Reason R: Water is present in large excess during hydrolysis.

In the light of above statements, choose the **correct** answer from the options given below:

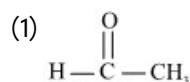
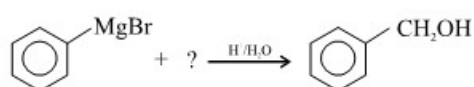
(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

Q66 What is the missing compound in the following transformation?



(3) CH_3CH_2COOH

(4) $HCHO$

Q67 Amphoteric oxides of Mn and Cr are:

(1) MnO and CrO

(2) Mn_2O_3 and Cr_2O_3

(3) MnO_2 and Cr_2O_3

(4) Mn_2O_7 and CrO_3

Q68 The IUPAC name of the symmetrical ketone that is produced on oxidation of an alcohol, $C_5H_{11}OH$ is:

(1) Pentan-1-one.

(2) Pentan-2-one.

(3) Pentan-4-one.

(4) Pentan-3-one.

Q69 At 300 K, for a reaction ΔH° is -66.9 kJ mol^{-1} and ΔG° is -41.8 kJ mol^{-1} . ΔG° for the same reaction at 330K is:

(1) -25.1 kJ mol^{-1}

(2) $+25.1$ kJ mol^{-1}

(3) $+18.7$ kJ mol^{-1}

(4) -39.3 kJ mol^{-1}

Q70 Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Ti^{3+} is a coloured ion.

Reason R: Ti shows +2, +3, +4 oxidation states.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

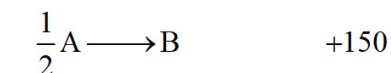
(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

Q71 Consider the following processes:



$\Delta H(\text{kJ/mol})$ 

For $B + D \longrightarrow E + 2C$, ΔH will be:

- (1) 525 kJ/mol
- (2) -175 kJ/mol
- (3) -325 kJ/mol
- (4) 325 kJ/mol

Q72 Which of the following statements is **incorrect**?

- (1) α -Black phosphorus is formed by heating of red phosphorus.
- (2) β -Black phosphorus does not burn in air upto 875 K.
- (3) White phosphorus readily catches fire in air to give dense fumes of P_4O_{10} .
- (4) Red phosphorus does not react with caustic alkalis.

Q73 Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Chemistry of actinoids is more complicated than lanthanoids.

Reason R: Actinoid elements are radioactive.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

Q74 α and β -forms of sulphur are at equilibrium at a temperature known as:

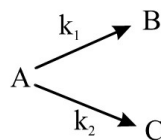
- (1) critical temperature.
- (2) transition temperature.
- (3) Boyle's temperature.
- (4) inversion temperature.

Q75 Salicylic acid on heating with sodalime forms:

- (1) phenol.
- (2) benzyl alcohol.
- (3) benzene.

(4) benzoic acid.

Q76 A substance undergoes first order decomposition. The decomposition follows two parallel first order reaction having $k_1 = 1.26 \times 10^{-4} s^{-1}$, and $k_2 = 3.8 \times 10^{-5} s^{-1}$. The percentage distribution of B and C are:



- (1) 75% B and 25% C
- (2) 80% B and 20% C
- (3) 60% B and 40 % C
- (4) 76.83% B and 23.17% C

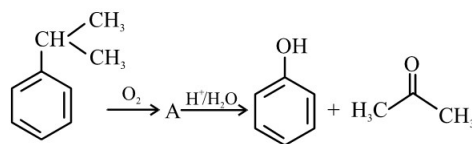
Q77 White phosphorus when boiled with strong solution of caustic soda produces:

- (1) sodium phosphide.
- (2) sodium phosphate.
- (3) phosphine.
- (4) red phosphorus.

Q78 Time required for 99.9% completion of a first order reaction is nearly:

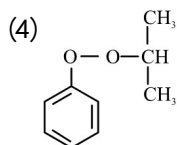
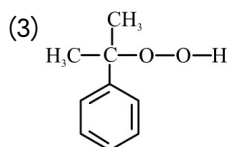
- (1) 20 times that required for 50% of the reaction.
- (2) 10 times that required for 50% of the reaction.
- (3) 25 times that required for 50% of the reaction.
- (4) 100 times that required for 50% of the reaction.

Q79 The structure of intermediate 'A' in the following reaction is:

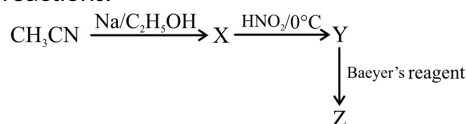


- (1)
- (2)





Q80 Identify the product Z in the given sequences of reactions:



- (1) CH_3CHO (2) CH_3CONH_2
(3) CH_3COOH (4) $\text{CH}_3\text{CH}_2\text{NHOH}$

Q81 Which of the following has largest atomic radius?

- (1) Sn (2) Si
(3) Pb (4) Ge

Q82 Which one of the following on treatment with 50% aqueous sodium hydroxide solution yields the corresponding alcohol and acid?

- (1) $\text{C}_6\text{H}_5\text{CHO}$ (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
(3) (4) $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$

Q83 How many moles of acidified FeSO_4 solution can be completely oxidised by one mole of KMnO_4 ?

- (1) 10 (2) 5
(3) 6 (4) 2

Q84 The final product by the treatment of glycerol with excess of HI is:

- (1) propane
(2) 2-iodopropane
(3) propene
(4) oxalic acid.

Q85 When 1.0 g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) is burned in a bomb calorimeter having heat capacity is 8.75 kJ, the temperature increases by 0.312 K. The enthalpy of combustion of oxalic acid at 27°C is:

- (1) -445.74 kJ/mol
(2) -644.45 kJ/mol
(3) -241.95 kJ/mol
(4) -346.94 kJ/mol

Q86 The activation energy of a reaction at a given temperature is found to be $2.303 RT \text{ Jmol}^{-1}$. The ratio of rate constant to the Arrhenius factor is:

- (1) 0.01 (2) 0.1
(3) 0.02 (4) 0.001

Q87 Which of the following diols would cleave into two fragments with HIO_4 ?

- (1) Hexan-1,3-diol
(2) Hexan-2,4-diol
(3) Hexan-1,6-diol
(4) Hexan-3,4-diol

Q88 Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Change in colour of acidic solution of potassium dichromate by breath is used to test drunken drivers.

Reason R: Change in colour is due to oxidising nature of alcohol with potassium dichromate.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

Q89 Which of the following acts as an oxidising agent?

- (1) B^{3+}
(2) Al^{3+}
(3) Tl^{3+}
(4) Both (A) and (C)

Q90 Transuranium elements are:

- (1) heavier than uranium.
(2) the part of the actinide series
(3) unstable and radioactive,
(4) All of these

Q91 For a first order gas phase reaction,
 $\text{A(g)} \rightarrow 2\text{B(g)} + \text{C(g)}$
 P_0 be the initial pressure of A and P_t is the total pressure at time 't'. The value of rate constant (k) is:

- (1) $\frac{2.303}{t} \log \left(\frac{P_0}{P_0 - P_t} \right)$



$$(2) \frac{2.303}{t} \log \left(\frac{2P_0}{3P_0 - P_t} \right)$$

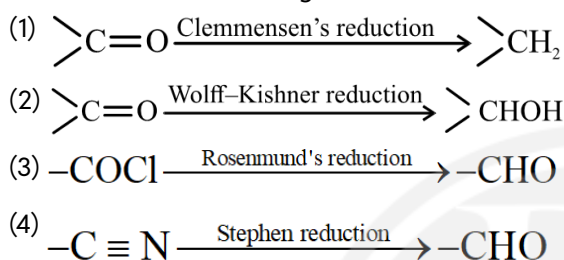
$$(3) \frac{2.303}{t} \log \left(\frac{P_0}{2P_0 - P_t} \right)$$

$$(4) \frac{2.303}{t} \log \left(\frac{2P_0}{2P_0 - P_t} \right)$$

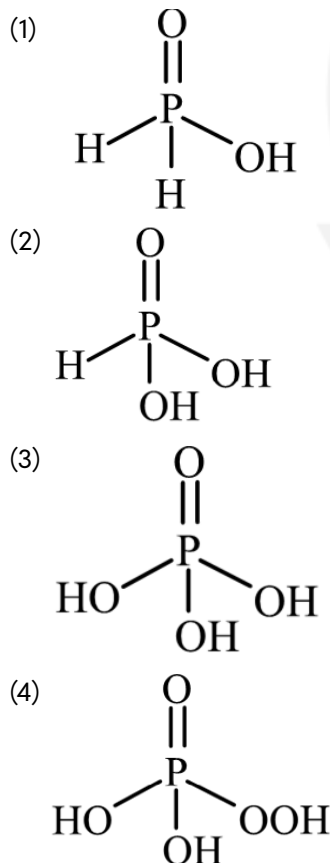
Q92 0.24 mole of phenol was treated with excess of bromine in the presence of water. All the phenol got quantitatively converted to bromophenol. The number of moles of Br_2 reacted were:

- (1) 0.24 mol. (2) 0.36 mol.
(3) 0.72 mol. (4) 0.48 mol.

Q93 Which one of the following reaction is **incorrect**?



Q94 The structural formula of orthophosphoric acid is:



Q95 The electronic configuration of element livermorium is:

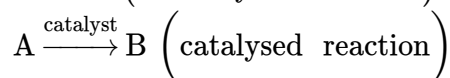
- (1) $[\text{Xe}] 4f^{14} 5d^{10} 6s^2 6p^4$

$$(2) [\text{Rn}] 5f^{14} 6d^{10} 7s^2 7p^4$$

$$(3) [\text{Kr}] 4d^{14} 5s^2 5p^4$$

$$(4) [\text{Xe}] 4f^{14} 5d^{10} 6s^2 6p^5$$

Q96 Consider the following reactions at 300 K.



The activation energy is lowered by $8.314 \text{ kJ mol}^{-1}$ for the catalysed reaction. How many times the rate of this catalysed reaction is greater than that of uncatalysed reaction? (Given $e^{3.33} = 28$)

- (1) 15 times (2) 38 times
(3) 22 times (4) 28 times

Q97 α -Halogenation of carboxylic acid is called:

- (1) Gattermann reaction
(2) Riemer-Tiemann reaction
(3) Sandmeyer's reaction
(4) HVZ reaction.

Q98 Given below are two statements:

Statement I: $\text{La}(\text{OH})_3$ is more basic than $\text{Lu}(\text{OH})_3$.

Statement II: Size of lanthanoids increases along the series and shows more covalent character.

In the light of above statements, Choose the **correct** option given below:

- (1) Both statement I and Statement II are correct.
(2) Both statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

Q99 Reactions of cyclohexanone with dimethyl amine in the presence of catalytic amount of an acid forms a compound if water during the reaction is continuously removed. The compound formed is generally known as:

- (1) Schiff's base.
(2) Enamine.
(3) Imine.
(4) Amine.

Q100 The four compounds are given below:

- I. CH_3OH
II. CH_3CHO
III. $\text{CH}_3\text{COC}_2\text{H}_5$



IV. C_6H_5CHO

In case of above compounds, which one of the following statements is **correct**.

- (1) I, III and IV undergo Aldol condensation.
- (2) II and III give Cannizzaro reaction.
- (3) II and III give Aldol condensation.
- (4) only II gives Haloform reaction.

Q101 All solids that settle from the _____ during primary treatment.

- (1) primary sludge
- (2) effluent
- (3) flocs
- (4) activated sludge

Q102 Identify from the following which is/are related to flower.

- I. Inflorescences are formed which bear the floral buds and then the flowers.
- II. Much before the actual flower is seen on a plant, the decision that the plant is going to flower has taken place.
- III. Androecium consists of a whorl of stamens.
- IV. Gynoecium represents the female gamete.

Choose the most appropriate answer from the options given below:

- (1) I, II, III and IV are correct
- (2) I, II and III are incorrect only
- (3) I, II and III are correct only
- (4) II and IV are incorrect only

Q103 Queen Victoria of England was a carrier for haemophilia. Since this is an X-linked disease, it can be predicted that:

- (1) all her sons would have that disease.
- (2) all her daughter would have been carriers.
- (3) Both her parent definitely had haemophilia.
- (4) haemophilia would have occurred in more of her male than her female descendants.

Q104 Read the following statements and mark the **correct** option.

- I. Mendel published his work on inheritance in 1865.
- II. By 1902, chromosome movement during meiosis had been worked out.
- III. Sutton united the knowledge of chromosomal segregation with mendelian principle and called it the chromosomal theory of inheritance.

IV. Morgan carried out several monohybrid crosses in *Drosophila* to study genes that were sex-linked.

Choose the most appropriate answer from the options given below:

- (1) I and III only
- (2) II and IV only
- (3) I and IV only
- (4) I, II and III only

Q105 Phenotypic and genotypic ratio is equal in:

- (1) $TTRR \times TtRr$.
- (2) $TtRR \times ttrr$.
- (3) $TtRr \times ttrr$.
- (4) $ttrr \times TTRr$.

Q106 Select the **mismatched** pair.

(1)	Pleiotropy	Multiple phenotypic expression
(2)	Test cross	Cross with recessive parent
(3)	Polygenic inheritance	Blood group
(4)	Incomplete dominance	Starch synthesis in pea

Q107 Select the **incorrect** statements.

- I. Embryo develops at the chalazal end of the embryo sac.
- II. Very less zygotes divide only after certain amount of endosperm is formed.
- III. The early stages of embryo development (embryogeny) are different in both monocotyledons and dicotyledons.
- IV. Embryo develops where the zygote is situated.

Choose the most appropriate answer from the options given below:

- (1) I and III only
- (2) II and IV only
- (3) I and IV only
- (4) I, II and III only

Q108 Select the **correct** for artificial hybridisation.

- (1) Emasculated flowers have to be covered with a bag of suitable colour.
- (2) Emasculation is done in female flowers.
- (3) Bagging is done with butter paper.



- (4) In unisexual flowers the female flower buds are bagged after the flowers open.

Q109 Select the **mismatched** pair.

(1)	Largest cell of the mature embryo sac	Central cell
(2)	Number of embryo sac in an ovule generally	Eight
(3)	Integumented megasporangium	Ovule
(4)	Nutritive tissue present in the ovule	Nucellus

Q110 Match **List-I** with **List-II**:

	List-I		List-II
(A)	Incomplete dominance	(I)	<i>Drosophila</i>
(B)	Linkage	(II)	<i>Antirrhinum</i> sp.
(C)	Sex linked recessive disorder	(III)	Sickle cell anaemia
(D)	Autosomal recessive disorder	(IV)	Colour blindness

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
 (2) A-IV, B-III, C-II, D-I
 (3) A-III, B-IV, C-II, D-I
 (4) A-II, B-I, C-IV, D-III

Q111 In Morgan's experiment, white and yellow genes were tightly linked, while white and miniature wing were loosely linked. The percent recombination shown by these genes were:

- (1) 50% each.
 (2) 72% and 8.3%, respectively.
 (3) 0.3% and 53%, respectively.
 (4) 1.3% and 37.2%, respectively.

Q112 Down's syndrome is an example of:

- (1) polyploidy. (2) aneuploidy.
 (3) deletion. (4) duplication.

Q113 Given below are two statements.

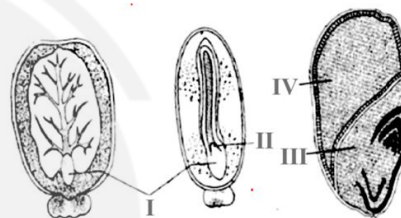
Statement I: Toddy, a traditional drink of some parts of southern India.

Statement II: Toddy is made by fermenting sap from bamboo.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect.
 (2) Statement-I is incorrect and Statement-II is correct.
 (3) Both Statement-I and Statement-II are correct.
 (4) Both Statement-I and Statement-II are incorrect.

Q114 In following three diagram identify **I, II, III** and **IV** respectively and find out the **correct** match among the following set of options:



	I	II	III	IV
(1)	Shoot apical meristem	Hypocotyl root axis	Scutellum	Endosperm
(2)	Hypocotyl root axis	Shoot apical meristem	Scutellum	Endosperm
(3)	Scutellum	Hypocotyl root axis	Shoot apical meristem	Endosperm
(4)	Endosperm	Scutellum	Hypocotyl root axis	Shoot apical meristem

Q115 Which of the following is **not** the feature of exine?

- (1) Made up of resistant organic material.
 (2) Germ pore is present.
 (3) Discontinuous layer.
 (4) Prominent sporopollenin deposition on germ pores.

Q116 Given below are two statements:



Statement I: The number and length of stamens are variable in flowers of different species.

Statement II: The bilobed nature of an anther is very distinct in the longitudinal section of the anther.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q117 The diploid and triploid product of double fertilization respectively are:

- (1) zygote and primary endosperm nucleus.
- (2) endosperm and cotyledons.
- (3) embryo and perisperm.
- (4) zygote and scutellum.

Q118 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Mendel proposed a third set of generalisations that we call Mendel's Law of Independent Assortment.

Reason R: When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q119 Rediscovery of Mendelian principle is associated with all scientists, **except**:

- | | |
|--------------|-------------------|
| (1) de Vries | (2) von Tschermak |
| (3) Correns | (4) Henking |

Q120 Given below are two statements.

Statement I: Majority of insect-pollinated flowers have large feathery stigma to trap pollen.

Statement II: Pollination guarantees the transfer of the right type of pollen.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct and Statement II is incorrect.
- (2) Statement I is incorrect and Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q121 The two-celled stage of mature pollen grain consists of:

- (1) vegetative cell and generative cell.
- (2) vegetative cell and one male gamete.
- (3) two male gametes.
- (4) generative cell and one male gamete.

Q122 Read the following statements:

I. Mendelian disorders are mainly determined by alteration or mutation in all gene.

II. The pattern of inheritance of such Mendelian disorders can be traced in a family by the pedigree analysis.

III. Colour blindness occurs in about 10 per cent of males and only about 4 per cent of females.

IV. Males have only one X chromosome and females have two.

Choose the most appropriate answer from the options given below:

- (1) II and III are correct only
- (2) I, II and III are incorrect only
- (3) I, II and III are correct only
- (4) II and IV are correct only

Q123 Select the **correct** statement from the following.

- (1) Biogas is produced by the activity of aerobic bacteria on animal waste.
- (2) *Methanobacterium* is an aerobic bacterium found in rumen of cattle.
- (3) Biogas is pure methane.
- (4) Activated sludge is sediment present in settlement tank of sewage treatment plant which is a right source of aerobic bacteria.

Q124 'Swiss cheese' bears large holes due to the production of CO₂ by which microbe?



- (1) *Lactobacillus*
 (2) *Saccharomyces cerevisiae*
 (3) *Propionibacterium sharmanii*
 (4) *Aspergillus niger*

Q125 Which of the following is **wrongly** matched in the given table?

	Microbe	Product
(1)	<i>Trichoderma polysporum</i>	Cyclosporin A
(2)	<i>Monascus purpureus</i>	Statins
(3)	<i>Streptococcus</i>	Streptokinase
(4)	<i>Clostridium butylicum</i>	Lipase

Q126 Select the **incorrect** statements.

- I. Bt-cotton is being cultivated in all states of our country.
 II. LAB produce acids that coagulate and completely digest the milk proteins.
 III. *Nucleopolyhedrovirus* shows effect on birds and fishes.
 IV. LAB improves milk nutritional quality by increasing vitamin B₁₂.

Choose the most appropriate answer from the options given below:

- (1) I, II and III only
 (2) II, III and IV only
 (3) I, III and IV only
 (4) II and III only

Q127 Match List-I with List-II:

List-I	List-II
(A) 	(I) Consanguineous mating
(B) 	(II) Sex unspecified
(C) 	(III) Five unaffected offspring
(D) 	(IV) Normal individuals

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
 (2) A-I, B-III, C-II, D-IV
 (3) A-II, B-IV, C-III, D-I
 (4) A-I, B-II, C-III, D-IV

Q128 Select the **incorrect** statements.

- I. Penicillin was extensively used to treat American soldiers wounded in World War I.
 II. Fleming, Chain and Florey were awarded the Nobel Prize in 1946 for the discovery of penicillin.
 III. Full potential of penicillin as an effective antibiotic was established by Alexander Flemming.
 IV. First antibiotic was discovered by Alexander Flemming.

Choose the most appropriate answer from the options given below:

- (1) I, II and III only
 (2) I, II and IV only
 (3) I and III only
 (4) III only

Q129 *Trichoderma* used to control plant disease is:

- (1) fungus. (2) bacteria.
 (3) arthropod. (4) virus.

Q130 Which of the following is known as 'clot-buster'?

- (1) Statins (2) Butyric acid
 (3) Streptokinase (4) Pectinase

Q131 Which of the following are used to ripen 'Roquefort cheese'?

- (1) Bacteria (2) Yeast
 (3) Fungi (4) Cyanobacteria

Q132 Select the **correct** option for the plants pollinated by water and its feature.

- (1) In *Vallisneria* the female flower reaches the surface of water.
 (2) In *Zostera* the female flower reaches the surface of water.
 (3) In Water lily the female flowers remain submerged in water.
 (4) In Water hyacinth the female flowers remain submerged in water.

Q133 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Mendel's experiments had small sample size which gave greater credibility to the data.

Reason R: A true breeding line shows a stable trait inheritance and expression for several



generations.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q134 Distillation of the fermented broth (which is formed by yeast) results in the formation of the following beverages:

- (1) whisky, brandy and beer.
- (2) whisky, wine and rum.
- (3) whisky, brandy and rum.
- (4) whisky, beer and rum.

Q135 How many of the following are **correct** for the blood group denoted by following genotype.

	Genotype of offspring	Blood types of offspring
A.	$I^A I^A$	A
B.	$I^A i$	A
C.	$I^A I^B$	AB
D.	$I^B i$	B
E.	II	O

Choose the most appropriate answer from the options given below:

- (1) Two
- (2) Four
- (3) Five
- (4) One

Q136 Select **correct** for *Azospirillum*.

- (1) It is cyanobacteria.
- (2) It fixes nitrogen in paddy fields.
- (3) It is free living.
- (4) It is used in controlling fungal diseases.

Q137 During Mendel's investigation, it was first time that **X** and **Y** were applied in biology. Here **X** and **Y** refers to:

- (1) X-statistical analysis; Y-mathematical logic
- (2) X-statistical analysis; Y-physical logic
- (3) X-statistical analysis; Y-chemistry logic
- (4) X-statistical analysis; Y-simple logic

Q138 Match **List-I** with **List-II**:

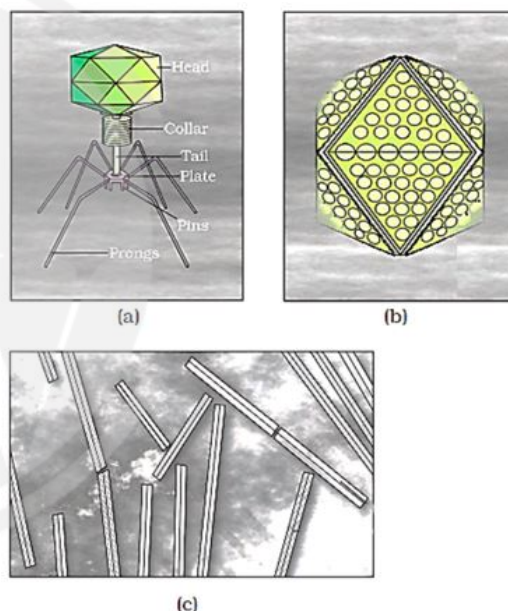
List-I	List-II
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(A) Cyclosporin-A	(I)	Clarifying bottles
(B) Proteases	(II)	Immunosuppressive agent
(C) Streptokinase	(III)	Myocardial infarction
(D) Lipases	(IV)	Oily stains

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-II, B-I, C-III, D-IV

Q139 Given below are diagrams of some viruses. Identify **a**, **b** and **c** and select the **correct** option for **a**, **b** and **c**.



- (1) a-Bacteriophage; b-Adenovirus; c-Tobacco Mosaic Virus (TMV).
- (2) a-Adenovirus; b-Tobacco Mosaic Virus (TMV); c-Bacteriophage
- (3) a-Tobacco Mosaic Virus (TMV); b-Bacteriophage; c-Adenovirus
- (4) a-Tobacco Mosaic Virus (TMV); b-Adenovirus; c-Bacteriophage

Q140 The secondary treatment of sewage waste is also known as:

- (1) chemical treatment.
- (2) physical treatment.
- (3) biological treatment.



(4) detoxification.

Q141 What are the chances of appearance of red flowered plants in a cross between pink and white flowered plant in dog flower?

- (1) Zero (2) 0.25
(3) 0.5 (4) 0.75

Q142 Person affected from _____ has total 45 chromosomes in each cell.

- (1) phenylketonuria
(2) Klinefelter's syndrome
(3) Down's syndrome
(4) Turner's syndrome

Q143 Identify from the following that can maintain long viability for pollen grains.

- (1) Rosaceae family and wheat
(2) Rice and Leguminosae family
(3) Leguminosae family, rice and wheat
(4) Solanaceae family

Q144 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Flowering plants have developed many devices to discourage cross pollination and to encourage self-pollination.

Reason R: Continued self-pollination result in inbreeding depression.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q145 Dragonflies can be used to get rid of how many of the following pests?

- I. Aphids
II. Butterfly caterpillars
III. Mosquitoes
IV. Caterpillars

Choose the most appropriate answer from the options given below:

- (1) One (2) Two
(3) Three (4) Four

Q146 α thalassemia and β thalassemia in humans is controlled by:

- (1) genes on chromosome 16 and 11 respectively.
(2) genes on chromosome 21 and 13 respectively.
(3) genes on chromosome 6 and 16 respectively.
(4) genes on chromosome 16 and 6 respectively.

Q147 Given below are two statements:

Statement I: Biocontrol refers to the use of biological methods for controlling plant diseases and pests.

Statement II: In agriculture, there is a method of controlling pests that relies on natural predation rather than introduced chemicals.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect.
(2) Statement-I is incorrect and Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

Q148 Select the **correct** statements related to Bt spores used as biocontrol agent.

- I. Mixed with organic solvent.
II. Available as sachets.
III. Sprayed onto vulnerable plants such as brassicas and fruit trees.
IV. On the skin of the larvae, the toxin is released and the larvae get killed.

Choose the most appropriate answer from the options given below:

- (1) I and III only
(2) II and IV only
(3) I and IV only
(4) II and III only

Q149 Select the **incorrect** statements for Down's syndrome.

- I. Psychomotor development is retarded.
II. Short statured
III. Large round head
IV. Gynaecomastia
V. Completely open mouth



Choose the most appropriate answer from the options given below:

- (1) I and II only
- (2) I, II and IV only
- (3) III, IV and V only
- (4) IV only

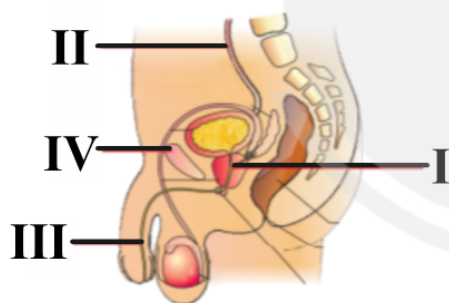
Q150 Size of typical pollen grain is:

- (1) 10 – 15 micrometers in diameter.
- (2) 25 – 50 micrometers in diameter.
- (3) 15 micrometers – 500 mm in width.
- (4) 100 nm – 50 micrometers in radius.

Q151 When more individuals acquire peripheral character value at both ends of the distribution curve, this type of natural selection is called:

- (1) stabilising selection.
- (2) directional selection.
- (3) disruptive selection.
- (4) none of these

Q152 Given below is a diagrammatic sectional view of male reproductive system. Identify a duct which is formed by the fusion of duct from seminal vesicle and vas deferens.



- (1) I
- (2) II
- (3) IV
- (4) III

Q153 Which of the following methods of contraception has maximum chances of failure?

- (1) Lactational amenorrhea
- (2) Vasectomy
- (3) Condoms
- (4) IUDs

Q154 In which following part of the female reproductive system does the secondary polar body is released after the completion of meiotic-II division?

- (1) Vagina
- (2) Oviduct

(3) Uterus

(4) Cervix

Q155 Choose the **incorrect** match w.r.t placental mammals and Australian marsupial showing convergent evolution.

- (1) **Placental mammals**- Flying squirrel, **Australian marsupials**-Flying phalanger
- (2) **Placental mammals**- Anteater, **Australian marsupials**-Numbat
- (3) **Placental mammals**- Lemur, **Australian marsupials**-Bobcat
- (4) **Placental mammals**- Wolf , **Australian marsupials**-Tasmanian wolf

Q156 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The edges of the isthmus possess finger-like projections called fimbriae.

Reason R: Fimbriae help in collection of the ovum after ovulation.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q157 In a human female, menstrual cycle is 40 days long, ovulation is expected occur on the;

- (1) 16th day.
- (2) 20th day.
- (3) 26th day.
- (4) 30th day.

Q158 Who proposed that 'the first form of life could have come from pre-existing non-living organic molecules'?

- (1) Oparin & Haldane
- (2) Stanley Miller
- (3) Lamarck
- (4) Hugo deVries

Q159 All of the following statements are correct, **except:**

- (1) Sperms are released from the seminiferous tubules by the process called spermiation.



- (2) Antrum is a fluid filled cavity present in the primary follicles.
 (3) Relaxin is secreted from the placenta.
 (4) Colostrum is absolutely essential to develop resistance for the new-born babies.

Q160 According to the special creation theory, the age of earth is about:

- (1) 8000 years. (2) 4000 years.
 (3) 2000 years. (4) 10000 years.

Q161 Given below are two statements.

Statement I: Cervical caps are inserted into the female reproductive tract to cover the uterus during coitus.

Statement II: The hormone releasing IUDs make the uterus suitable for implantation.

In the light above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

Q162 According to Hugo deVries, mutations are:

- (1) small and directionless.
 (2) random and directional.
 (3) random and directionless.
 (4) small and directional.

Q163 Select the **correct** sequence of embryonic development.

- (1) Morula → Zygote → Blastocyst
 (2) Blastocyst → Morula → Zygote
 (3) Zygote → Morula → Blastocyst
 (4) Morula → Blastocyst → Zygote

Q164 In India, population crossed one billion mark in 2000. The probable reasons for this are, decline in:

- I. maternal mortality rate.
 II. infant mortality rate.
 III. number of people in reproductive age.
 IV. death rate.

Choose the most appropriate answer from the options given below.

- (1) I and II only
 (2) I, II and III only
 (3) I, II and IV only
 (4) I, II, III and IV

Q165 Saltation indicates:

- (1) single step small mutation.
 (2) single step large mutation.
 (3) multiple steps large mutation.
 (4) multiple steps small mutation.

Q166 Which of the following is a hormone releasing intra-uterine device (IUDs)?

- (1) Lactational amenorrhea
 (2) LNG - 20
 (3) Cervical caps
 (4) Cu7

Q167 Read the following statements (I-IV).

I. Intentional or voluntary termination of pregnancy before full term is called medical termination of pregnancy.

II. Genital warts is a non-sexually transmitted disease.

III. Pills have to be taken daily for a period of 21 days starting preferably within the first five days of menstrual cycle.

IV. In artificial insemination (AI) technique, the semen collected either from the husband or a healthy donor is artificially introduced into the vagina.

Which of the above statements is/are **incorrect**?

- (1) I and II only
 (2) I, II and IV only
 (3) III and IV only
 (4) II only

Q168 The female external genitalia include:

- I. ovary.
 II. mammary gland.
 III. mons pubis.
 IV. clitoris.
 V. labia majora.

Choose the most appropriate answer from the options given below.

- (1) I and II only
 (2) I and V only



- (3) III, IV and V only
(4) II, III and IV only

Q169 Spermicidal creams, jellies and foams are usually used along with the barrier methods to:

- (1) increase the conception efficiency.
(2) decrease their contraceptive efficiency.
(3) increase their contraceptive efficiency.
(4) provide motility to sperms.

Q170 Contraceptive pills used by females to prevent pregnancy are very effective with lesser side effects. They work by:

- (1) inhibiting gestation.
(2) inhibiting implantation.
(3) altering the quality of cervical mucus to prevent/retard the entry of sperms.
(4) both (B) and (C)

Q171 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: All copulations do not lead to fertilisation and pregnancy.

Reason R: Fertilisation can occur only if ovum and sperms are transported simultaneously to ampullary region.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q172 The extinct hominid who lived in near-east and central Asia between 1,00,000 - 40,000 years back and used hides to protect their body, were:

- (1) *Dryopithecus*.
(2) *Australopithecines*.
(3) *Homo habilis*.
(4) Neanderthal man.

Q173 Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Reproductive tract infection	(I)	Prevent sperms to reach cervix

(B)	Condom	(II)	Copper releasing IUDs
(C)	Tubectomy	(III)	Chlamydia
(D)	Multiload375	(IV)	Sterilisation in females

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-I, C-II, D-III

Q174 If frequency of a dominant allele in a population which is in genetic equilibrium is 0.7, then calculate frequency of heterozygotes in that population.

- (1) 0.21 (2) 0.24
(3) 0.48 (4) 0.42

Q175 Read the following statements (I-IV).

- I. It is a barrier method of contraception.
II. It covers the cervix during coitus.
III. It is made of rubber.
IV. It is reusable.

Which of the above statements are **correct** w.r.t vaults?

- (1) I and II only
(2) I, II and III only
(3) I, II and IV only
(4) I, II, III and IV

Q176 Which of the following is released from the ovary at the time of ovulation?

- (1) Primary oocyte
(2) Secondary oocyte
(3) Tertiary follicles
(4) Oogonia

Q177 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Sweet potato and potato are example of homologous organs.

Reason R: Sweet potato is root modification while potato is stem modification.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.



- (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

Q178 Which of the following is an **incorrect** statement w.r.t periodic abstinence?

- (1) The couple should abstain from coitus from day 10 to 17 of the menstrual cycle.
 (2) It does not have any side-effects.
 (3) It is a natural method of contraception.
 (4) Ovum and sperms are prevented from physically meeting with the help of barriers.

Q179 Which of the following is/are the key concepts of Darwinian theory of evolution?

- (1) Adaptive radiation
 (2) Natural selection
 (3) Branching descent
 (4) Both (B) and (C)

Q180 Given below are two statements.

Statement I: In ICSI (intra cytoplasmic sperm injection) procedure, sperms are directly injected into the cytoplasm of ovum to form an embryo.

Statement II: Infertility in male partner could be corrected by artificial insemination.

In the light above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

Q181 Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Fish like reptiles	(I)	18,000 years ago
(B)	Dagger like teeth	(II)	<i>Ichthyosaurs</i>
(C)	Pre-historic cave art	(III)	65 mya
(D)	Disappearance of Dinosaurs	(IV)	<i>Tyrannosaurus rex</i>

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
 (2) A-II, B-IV, C-I, D-III
 (3) A-III, B-I, C-IV, D-II
 (4) A-IV, B-I, C-II, D-III

Q182 All of the following statements about ZIFT are correct, but one is incorrect. Which of the following option is **incorrect**?

- (1) It is zygote intra fallopian transfer.
 (2) Early embryos upto 8 blastomeres can be transferred into the fallopian tube.
 (3) It is included in test tube baby method.
 (4) It involves *in-vivo* fertilisation.

Q183 In his experiments, S.L. Miller used which of the following in a closed flask?

- (1) C_2H_6 , O_2 , N_2 and water vapour at $800^\circ C$.
 (2) CH_4 , H_2 , NH_3 and water vapour at $600^\circ C$.
 (3) C_2H_6 , H_2 , NH_3 and water vapour at $600^\circ C$.
 (4) CH_4 , H_2 , NH_3 and water vapour at $800^\circ C$.

Q184 Which of the following options is **correct** w.r.t the role of oxytocin in the pregnant females?

- (1) It stimulates posterior pituitary to secrete vasopressin.
 (2) It causes strong uterine contractions during parturition.
 (3) It is secreted only by placenta.
 (4) It stimulates growth of mammary glands.

Q185 Read the following statements (I-V).

- I. Cleavage occurs as the zygote moves through isthmus towards fimbriae of infundibulum.
 II. In tertiary follicle, theca is organised into an inner theca interna and outer theca externa.
 III. Endometrium is an outermost glandular layer of uterus.
 IV. The mammary glands of the female undergo differentiation during pregnancy and starts producing milk towards the end of pregnancy by the process called lactation.
 V. Inner group of cells attached to trophoblast is called the inner cell mass.

Which of the above statements are **incorrect**?

- (1) I and III only
 (2) III and IV only



- (3) I, II, III and V only
(4) II, III, IV and V only

Q186 What is the difference in years of appearance of first non-cellular forms of life and appearance of first cellular forms of life?

- (1) 3000 million years
(2) 2000 million years
(3) 1000 million years
(4) 5000 million years

Q187 Given below are two statements.

Statement I: Coitus interruptus is natural contraceptive method in which the male partner withdraws his penis from the vagina just before ejaculation so as to avoid insemination.

Statement II: Saheli is a new oral contraceptive for the females was developed by scientists at Central Drug Research Institute (CDRI) in Lucknow, India.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q188 Select the **correct** option which includes all sexually transmitted diseases.

- (1) Cancer, AIDS, syphilis.
(2) Gonorrhoea, syphilis, genital herpes.
(3) Gonorrhoea, malaria, genital herpes.
(4) AIDS, malaria, Hepatitis-B.

Q189 According to Hardy-Weinberg equilibrium the tendency of population to remain in genetic equilibrium is affected by:

- (1) gene flow.
(2) lack of migration.
(3) lack of mutations.
(4) no genetic recombination.

Q190 How many technologies in the list given below perform *in-vivo* fertilisation?

GIFT, ICSI, IUT, AI, IUI

- (1) Five
(2) Three
(3) Two
(4) One

Q191 Read the following statements (I-V).

- I. In adults, each testis is oval in shape.
II. Each testis has about 150 compartments.
III. Each testicular lobule contains three to five highly coiled seminiferous tubules.
IV. The enlarged end of penis is called the glans penis.
V. Rete testis is a male sex accessory duct.

Which of the above statements are **incorrect**?

- (1) I, II and III only
(2) II and V only
(3) I, II, IV and V only
(4) II and III only

Q192 Which of the following is the main function of copper ions in copper releasing IUDs?

- (1) They alter the spermatogenesis.
(2) They suppress sperm motility.
(3) They make fallopian tube hostile to egg.
(4) They promote ovulation.

Q193 Read the following statements (I-IV).

- I. Fossils are remains of hard parts of life-forms found in rocks.
II. Jawless fish probably evolved around 350 mya.
III. The brain capacities of *Homo habilis* were between 650-800cc.
IV. The original Darwin finches were seed-eating.

Which of the above statements are **correct**?

- (1) I, II and III only
(2) I and II only
(3) I, II and IV only
(4) I, II, III and IV

Q194 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: During spermatogenesis, sperm heads are embedded in Sertoli cells.

Reason R: Sertoli cells are found outside seminiferous tubules.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.



- (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

Q195 How many spermatozoa and ova are produced from 100 secondary spermatocytes and 100 primary oocytes respectively?

- (1) 400 spermatozoa and 200 ova
 (2) 100 spermatozoa and 100 ova
 (3) 400 spermatozoa and 100 ova
 (4) 200 spermatozoa and 100 ova

Q196 Match the List I with List II.

List-I		List-II	
(A)	Heart sound	(I)	By the end of second month of pregnancy
(B)	Foetus develops limbs and digits	(II)	During the fifth month
(C)	Formation of major organ system	(III)	First sign of growing foetus
(D)	First movement and appearance of hair on head	(IV)	By the end of 12 weeks

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
 (2) A-IV, B-I, C-II, D-III
 (3) A-III, B-I, C-IV, D-II
 (4) A-II, B-III, C-I, D-IV

Q197 Given below are two statements:

Statement I: Gills of prawn and lungs of man are analogous organs.

Statement II: *Tyrannosaurus rex* was about 40 feet in height.

In the light of the above statements, choose the most appropriate answer from the options given below:

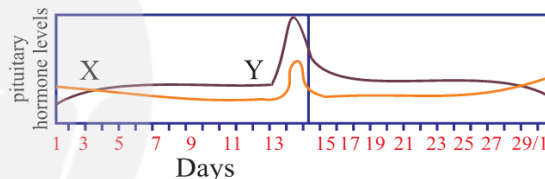
- (1) Statement I is correct but Statement II is incorrect.

- (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

Q198 All of the following statements are correct, **except:**

- (1) Vertebrate hearts and brains exhibit convergent evolution.
 (2) Fitness is the end result of the ability to adapt and get selected by nature.
 (3) A gene pool is total genes and their alleles in a population.
 (4) When more than one adaptive radiation appeared to have occurred in an isolated geographical area, it is called convergent evolution.

Q199 The following graph shows the level of hormones during a menstrual cycle. Choose the **correct** option.



- (1) X-LH, Y-FSH
 (2) X-Estrogen, Y-Progesterone
 (3) X-FSH, Y-LH
 (4) X-Progesterone, Y-Estrogen

Q200 The **correct** chronological order of human evolution from earliest to recent is:

- (1) *Australopithecines* → *Ramapithecus* → *Homo habilis* → *Homo erectus*
 (2) *Ramapithecus* → *Australopithecines* → *Homo habilis* → *Homo erectus*
 (3) *Ramapithecus* → *Homo habilis* → *Australopithecines* → *Homo erectus*
 (4) *Australopithecines* → *Homo habilis* → *Ramapithecus* → *Homo erectus*



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